

Exploratory Data Analysis - Singapore Total Fertility Rate & Total Live Births

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Introduction

This exploratory data analysis (EDA) investigates the temporal features of two significant population indicators for Singapore:

- **Total Fertility Rate (TFR):** the average number of children a woman would have over her lifetime at current age-specific fertility rates;
- **Total Live Births (TLB):** the total number of live births recognised per year.

The dates for the SingaporeCensus.csv reference dataset span from 1960 to 2024, sourced from the Singapore Department of Statistics. The objective is to recognise and analyse long-run trends, assess stationarity, and identify candidate time series models for a proposed training period 1960 - 2012 and a proposed forecasting performance assessed over the remaining 2013 - 2024 period.

Data Loading & Reshaping

The given data is in a CSV file, which is optimal for computation. However, the data is in a ‘wide format’, where its temporal variables (i.e. years) are given as columns instead of years. We transpose the data and extract the two data columns we are interested in.

Below is the transposition framework of the data.

Wide format (raw CSV)

DataSeries		1960		1961		...		2024
TFR row								
TLB row								

pivot_longer()

+

filter()

+

arrange()

Long format (processed)

Series		Year		Value
TFR		1960		5.76
TFR		1961		5.45
TLB		1960		61775
TLB		1961		62157

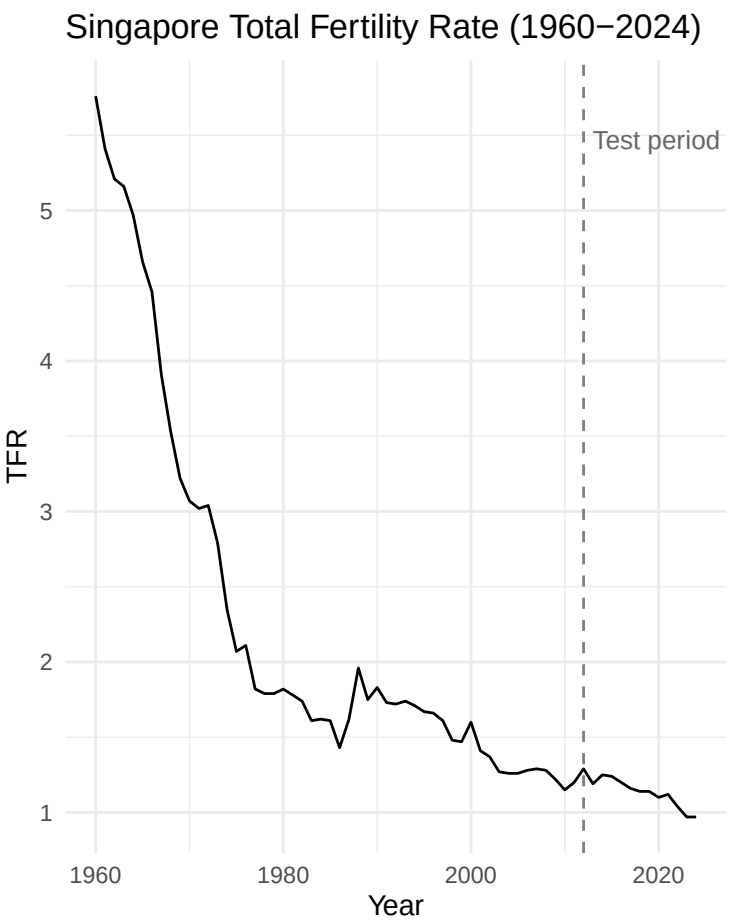
Data Quality & Summary

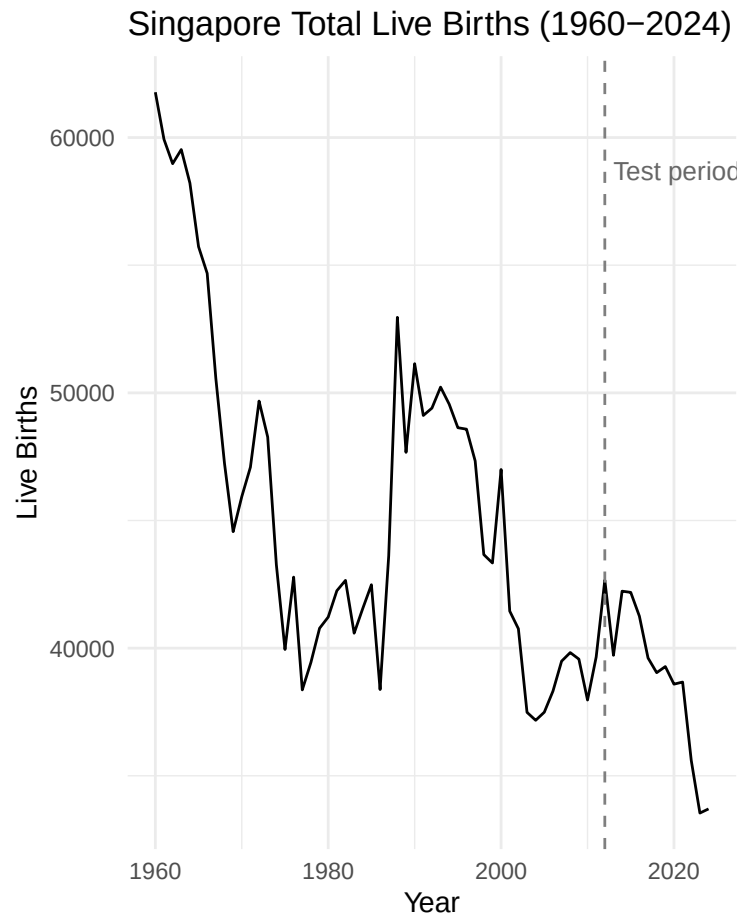
Table 1: Summary statistics for TFR and TLB (1960-2024)

Statistic	TFR	TLB
Min	0.97	33541.00
Q1	1.26	39570.00
Median	1.62	42484.00
Mean	2.06	44359.63
Q3	2.07	48577.00
Max	5.76	61775.00
NAs	0.00	0.00

Time Series Plots

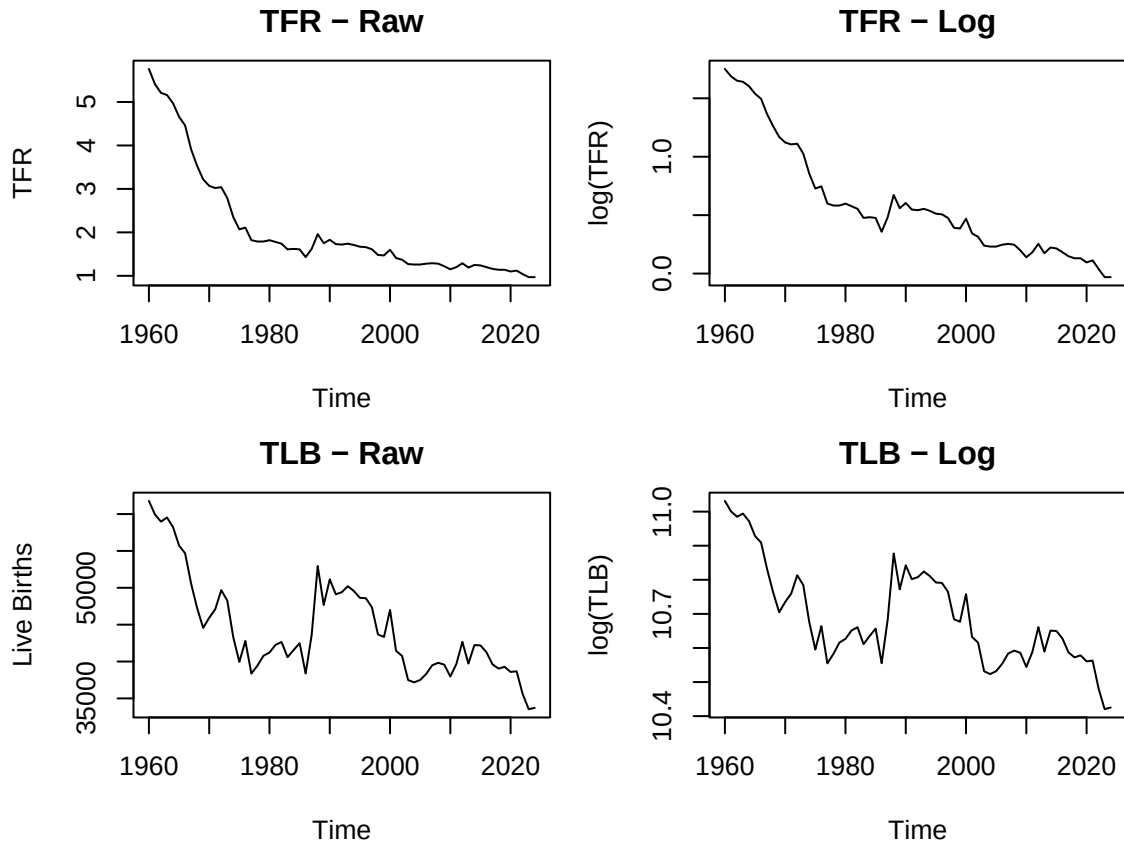
Below, the dashed vertical line marks the train-test split at 2012.





Temporal Feature Evaluation

Log Transformation Assessment



Stationarity Testing

Raw Series

Table 2: Stationarity tests - raw training series (1960-2012)

Series	ADF p-value	KPSS p-value
TFR	0.120	0.0100
TLB	0.356	0.0205

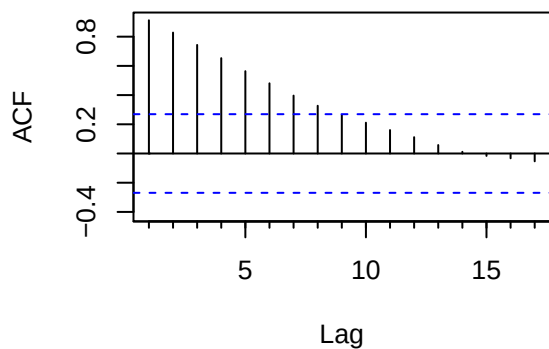
First Difference Series

Table 3: Stationarity tests - first-differenced training series

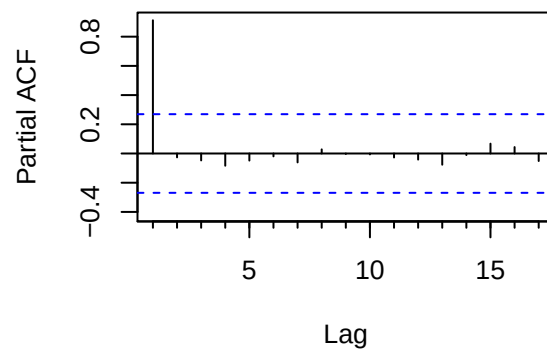
Series	ADF p-value	KPSS p-value
Δ TFR	0.0830	0.01
Δ TLB	0.0395	0.10

ACF & PACF Analysis

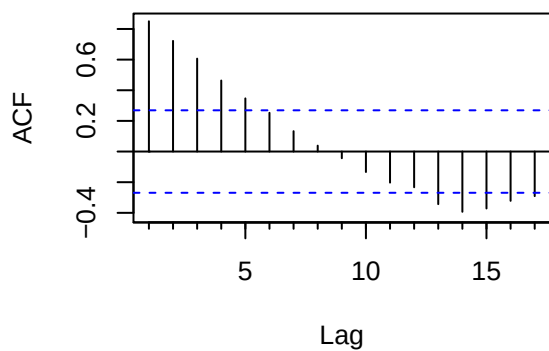
TFR - ACF



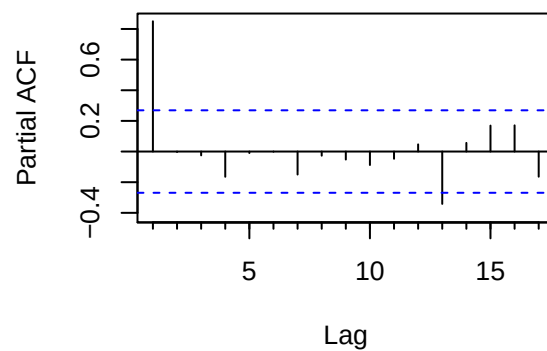
TFR - PACF

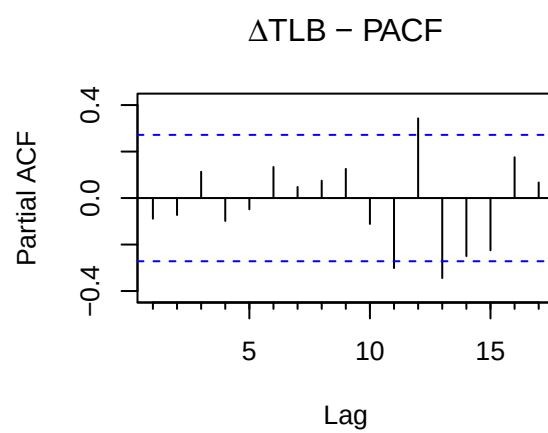
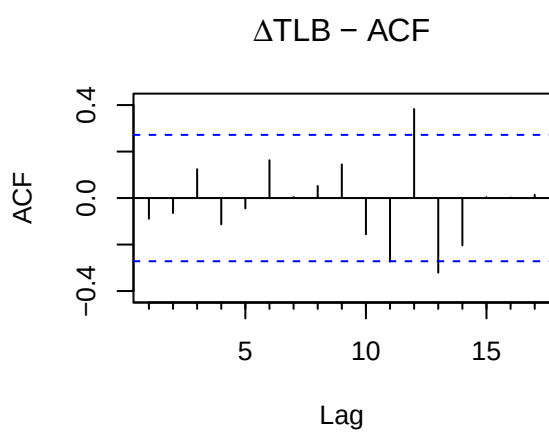
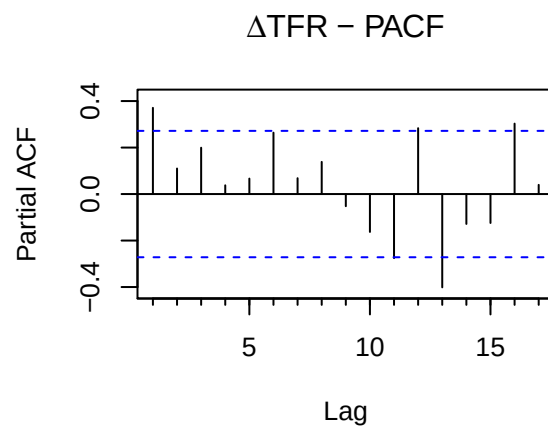
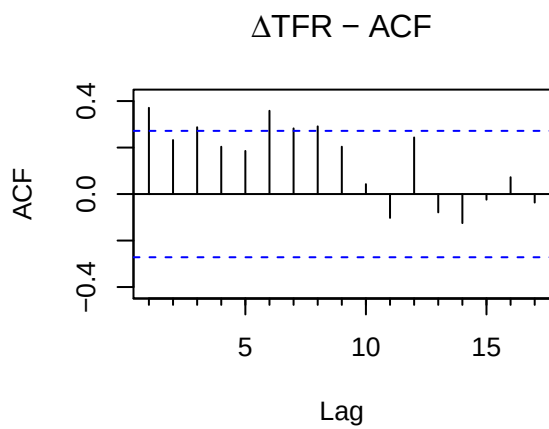


TLB - ACF

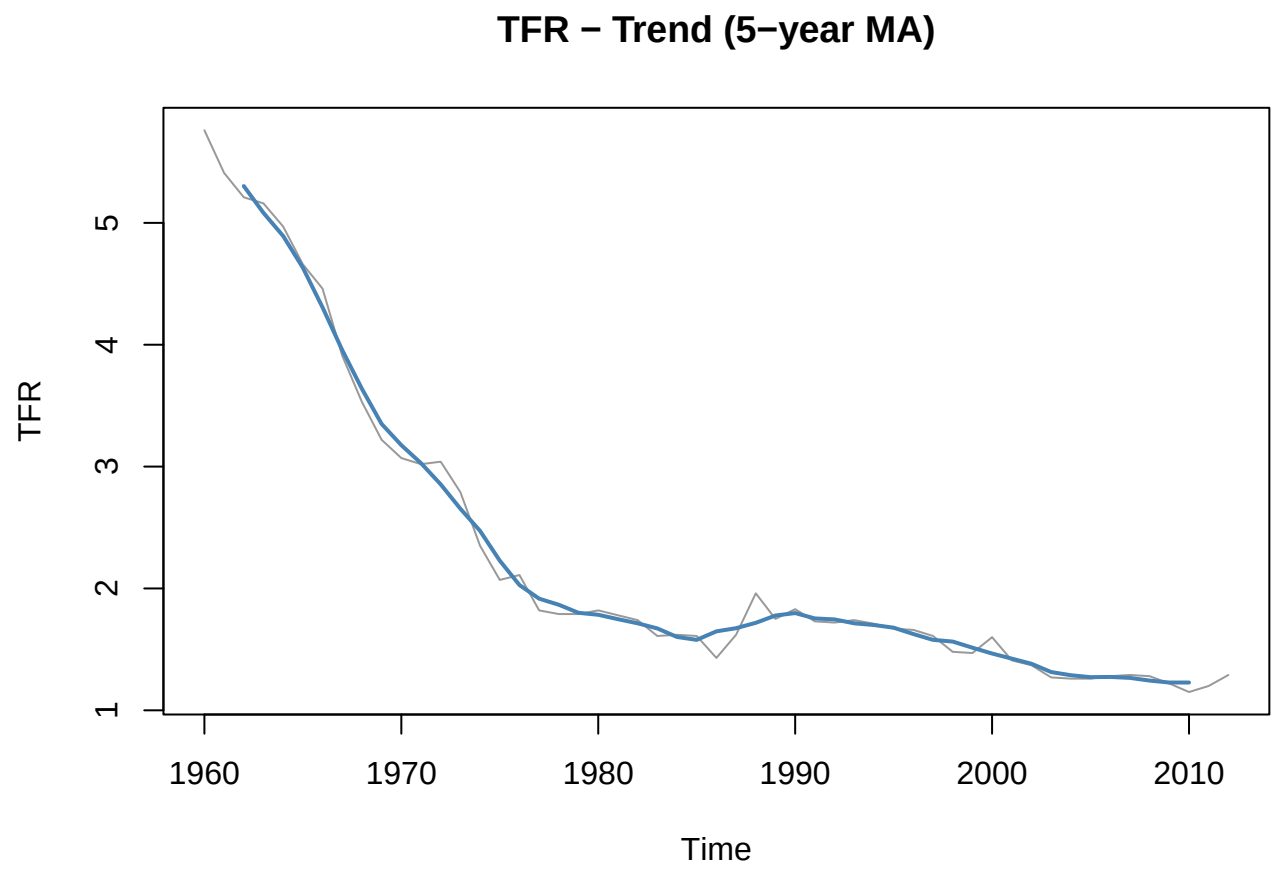


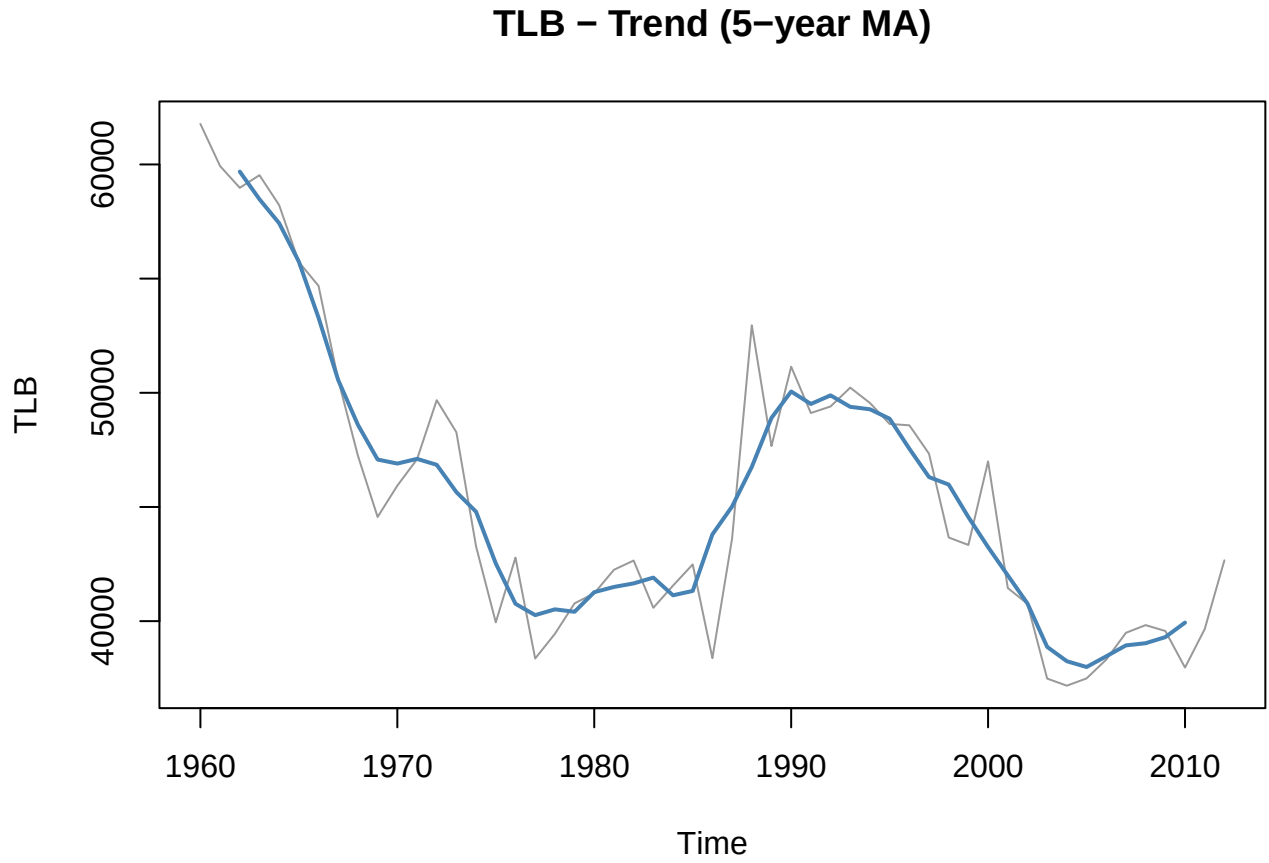
TLB - PACF





Decomposition





Candidate Model Fitting

TFR - Candidate Model

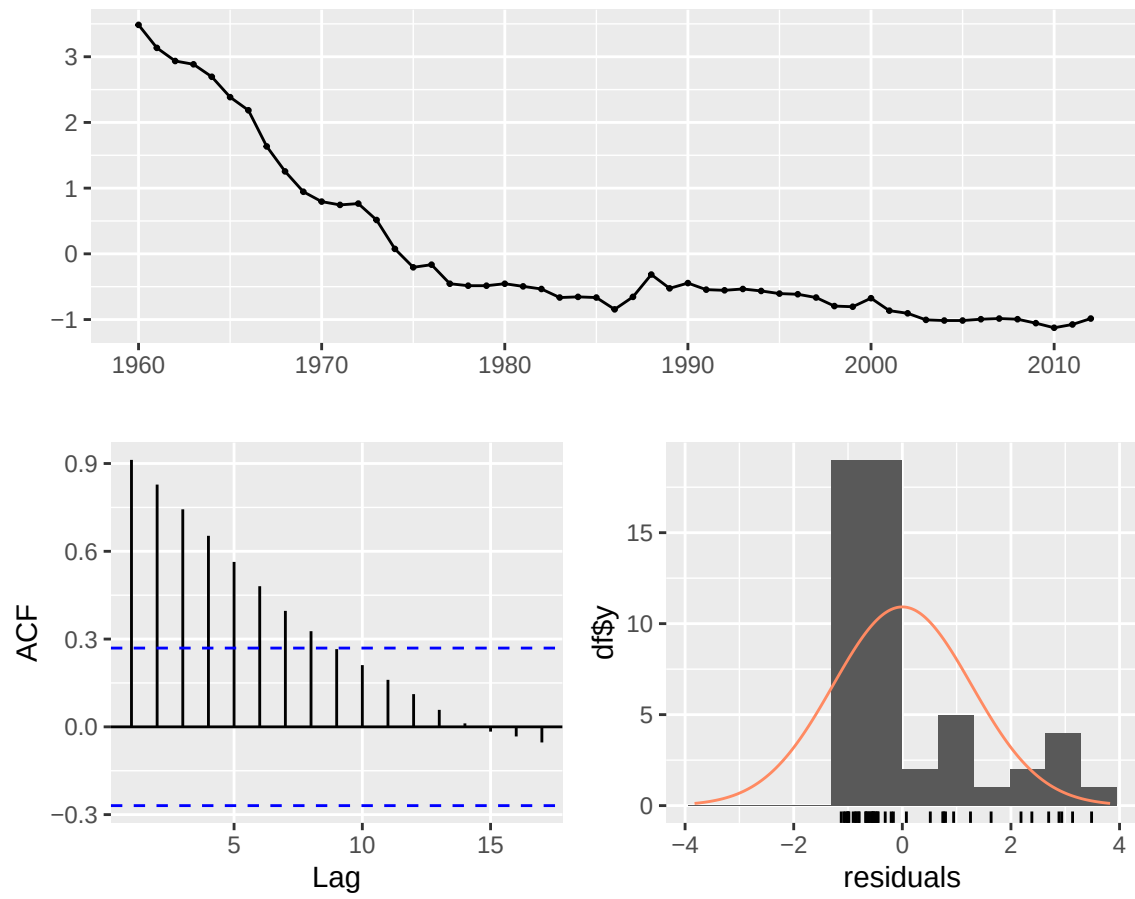
Table 4: TFR training set error measures

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0	1.2631	0.9975	-23.1514	45.3835	7.9433	0.9124

Table 5: TFR Ljung-Box test on residuals (lag = 10)

Q*	df	p-value
201.6392	10	0

Residuals from ARIMA(0,0,0) with non-zero mean



TLB - Candidate Model

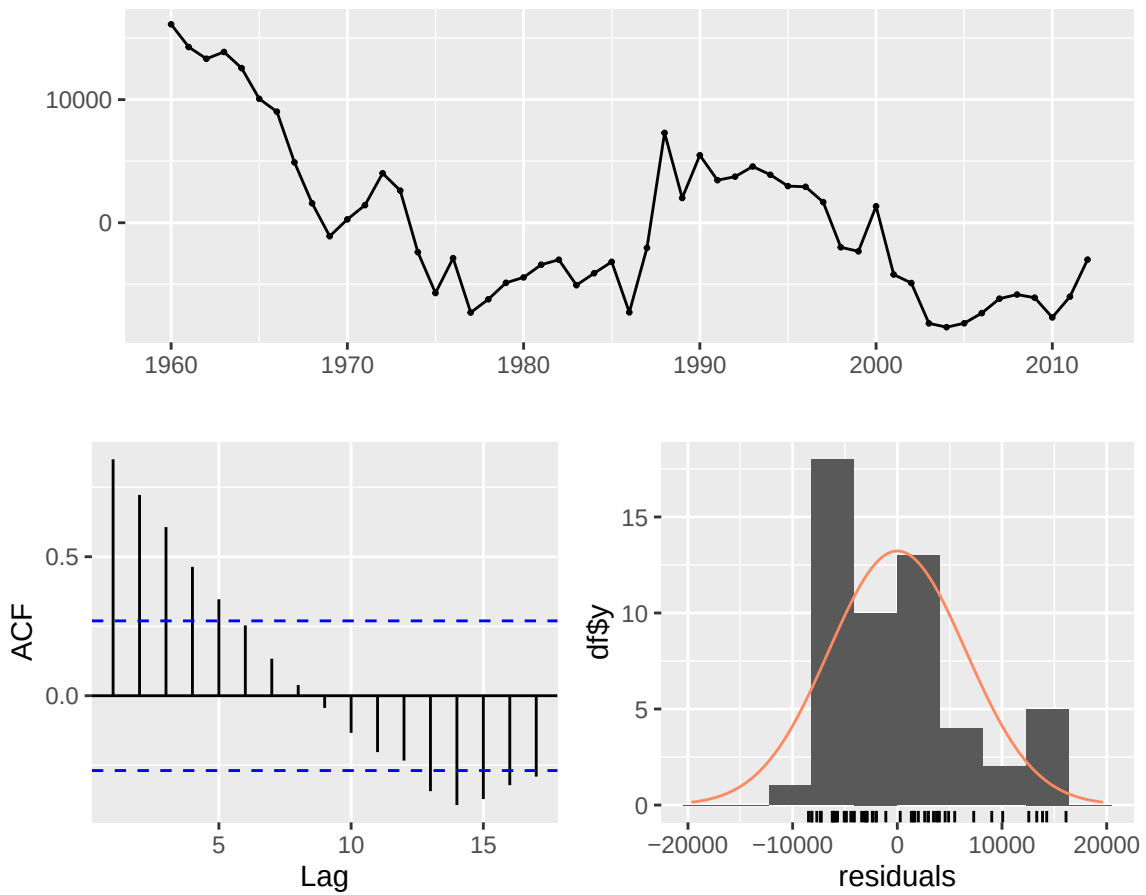
Table 6: TLB training set error measures

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0	6492.436	5410.766	-1.888	11.7319	2.5851	0.8504

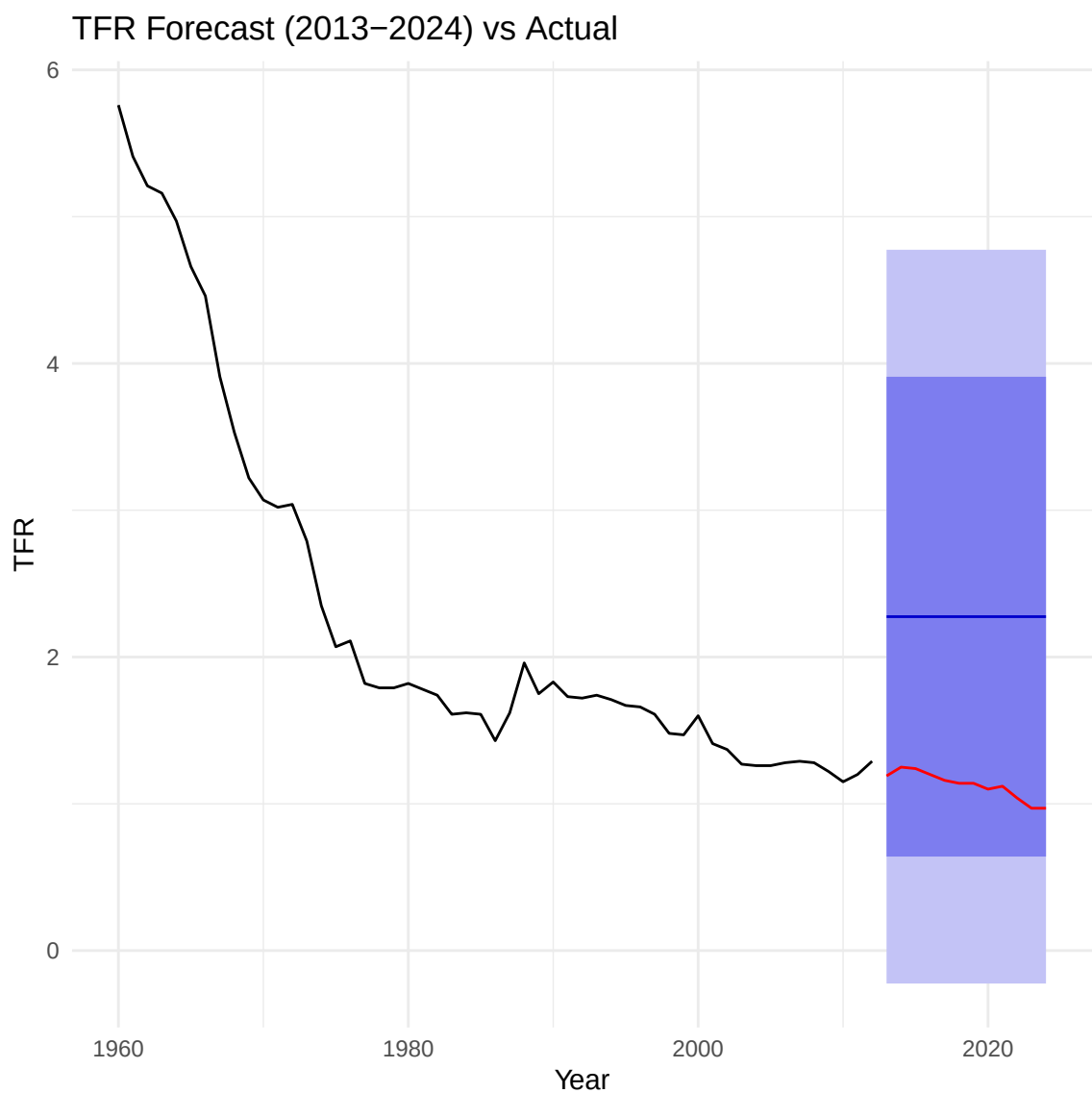
Table 7: TLB Ljung-Box test on residuals (lag = 10)

Q*	df	p-value
118.416	10	0

Residuals from ARIMA(0,0,0) with non-zero mean



Forecasting Model & Assessment



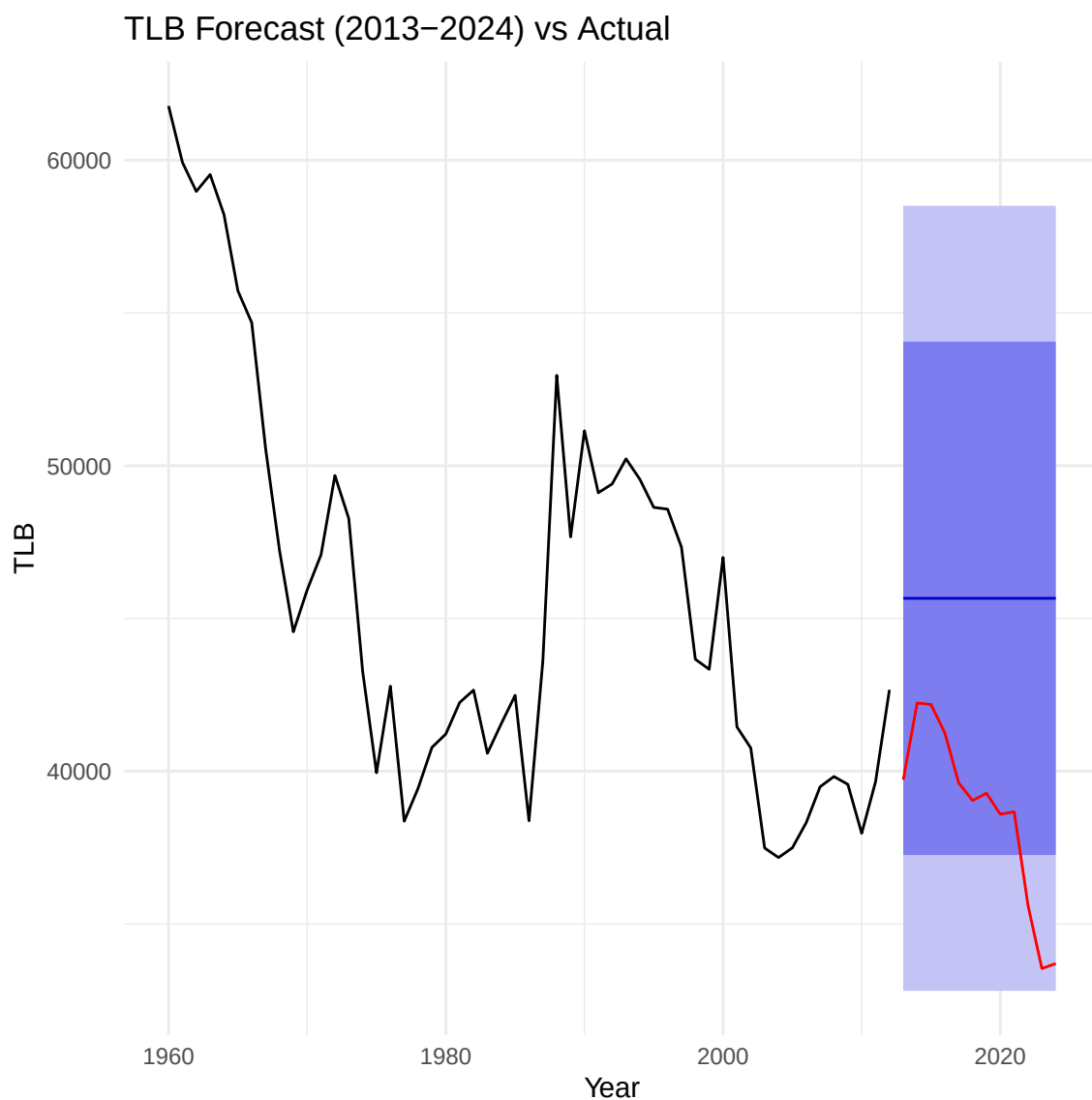


Table 8: TFR forecast accuracy

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.0000	1.2631	0.9975	-23.1514	45.3835	7.9433	0.9124	NA
Test set	-1.1484	1.1519	1.1484	-103.2781	103.2781	9.1452	0.7447	26.9001

Table 9: TLB forecast accuracy

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.000	6492.436	5410.766	-1.8880	11.7319	2.5851	0.8504	NA
Test set	-7039.987	7578.983	7039.987	-18.8901	18.8901	3.3635	0.7366	5.4094

Key Temporal Insights

The figures throughout this EDA show a sharp decline in both TLB and TFR post-1960s, which is indicative of Singapore's rapid demographic transformation. TLB drops from peaks around 1965 (~60,000 births) among the early baby booms, stabilising below 40,000 by the 2000s; TFR plummets from >4.5 in 1960 to a remaining fifth of this figure, with its notable troughs in the 1970s (during the presence of their two-limit policy) and 2020s (during the COVID-19 pandemic). Visual patterns indicate strong negative trends, potential cyclical behaviour, and increased volatility post-2010.

Data Quality

The given Singapore Census CSV data is objectively clean; no NAs present, so no discrete approximations are made. However, a limitation is the lack of monthly granularity for our seasonality tests. This could be an inviting avenue to investigate, if present.

Key Visual & Test Insights

Line plots are indicative of potential exponential decay (log-TLB linearises trend); differenced series highlight stationarity (expected ADF $p < 0.05$ post- $d=1$). ACF shows high persistence (lag-1 ~ 0.9), weak cycles around 10-15 years (implementation of national policies, likely). In decomposition, trend is massive (~90% variance), with minimal seasonality in annuals (see Data Quality for relational limitations).

Research Areas

1. What policy interventions over this sample period best explain the TFR/TLB deviations from smooth decline trend?
2. Do ARIMA/ETS models trained on the 1960-2012 slice accurately backcast 2013-2024 trends?
3. How do age-shifts and ethic gaps explain < 1.0 TFR?